Technology Stack

Date	30 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	HTML5, CSS3, JavaScript, Bootstrap	Provides a responsive and user-friendly interface for applicants to enter loan details and view eligibility results.
2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	Python 3.10+, Flask Framework	Lightweight Flask API controller handles web requests, user authentication sessions, and calls the ML model to predict loan eligibility.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	MySQL, SQLAlchemy ORM	Relational storage maps user details, loan applications, prediction history, and results securely.
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	Local server, Render / Heroku (Optional)	Used for development, testing, and deployment of the Smart Lender application.
5	**Version Control & CI/CD** *(e.g., GitHub, GitLab, Jenkins)*	Git, GitHub	Collaborative version control and branch merging.
6	**Machine Learning & Other Tools** *(e.g., Scikit-learn, Pandas, TensorFlow)*	Scikit-learn, Pandas, NumPy, Pickle, Matplotlib	Scikit-learn builds the loan eligibility prediction model, Pandas & NumPy handle data preprocessing, Pickle stores the trained model, and Matplotlib generates performance visualizations.